

Clinical Document Intelligence Hub

Automating Healthcare Document Processing

Problem Context

Healthcare providers spend thousands of hours manually reviewing unstructured clinical documents ? discharge summaries, lab reports, intake forms.

Core Bottlenecks

Manual data entry is slow, inconsistent, and highly error-prone ? creating administrative delays in patient care pathways.

Project Objective

Build a working proof-of-concept AI prototype that ingests unstructured clinical inputs (PDF, TXT, Image) and surfaces decision-ready structured intelligence.

Value Proposition

Transform fragmented document text into standardized clinical outputs, risk alerts, and evidence-backed next steps.

70%+

reduction in manual workload

Solution Architecture & Design Flow

Hybrid AI Ingestion, Extraction, and Safety Verification Architecture

1

Document Ingestion Layer

Multi-format parsing accepting PDFs (PyMuPDF), text files, and image scans.

2

Clinical LLM Extraction

Prompt-engineered structured extraction forcing strict Pydantic JSON schema adherence.

3

Deterministic Safety & Risk Engine

Rules engine post-processes extracted facts · e.g. WBC > 11.0, Penicillin + Amoxicillin allergy contraindications.

4

Confidence & Evidence Layer

Assigns confidence scores (0-100%) and extracts exact source-text evidence quotes for total clinical auditability.

5

Streamlit Dashboard UI

Clean clinical interface featuring Patient Overview Cards, Risk Flags, Diagnoses/Meds, Lab Tables, and JSON Export.

Implementation Highlights

Technical Decisions, AI Logic & System Quality Controls



Structured JSON Schema

Enforced using Pydantic models ? Patient, Medication, LabResult, RiskFlag, ClinicalSummary.



Hybrid Reasoning Architecture

LLM used for unstructured text extraction, paired with deterministic code rules for high-stakes medical thresholds.



Evidence Traceability

Every risk flag and abnormal lab includes exact quote citations from the source document.



Zero-Downtime Multi-Provider Strategy

Seamless support for OpenAI GPT-4o, Google Gemini 2.5, and a zero-dependency deterministic fallback engine.

Challenges & Learnings

Addressing Hallucinations, Safety Scoping, and Unstructured Data Variability

1

Hallucination Prevention

Strictly constrained LLM prompt rules to extract ONLY explicitly stated facts; unmentioned fields return null.

2

Clinical Safety

Scoped all recommendations as decision-support suggestions rather than autonomous diagnoses.

3

Document Formatting Variance

Standardized multi-format text extraction pipeline to handle noisy PDFs and plain text notes uniformly.

KEY TAKEAWAY

Combining LLM semantic flexibility with deterministic rule validation provides the ideal balance of accuracy and safety in healthcare software.

Demo Summary & Next Steps

Working Prototype Deliverables & Strategic Roadmap

Delivered Artifacts

Full working Streamlit application, sample datasets (PDF/TXT), and 1-page executive README.

Next Steps · Strategic Roadmap

1

Healthcare Integration

HL7 / FHIR API export for direct EHR integration ? Epic, Cerner, AthenaHealth.

2

Safety & Security

Enterprise HIPAA compliance, PHI de-identification (PII/PHI anonymization pipeline), and OAuth2 RBAC.

3

Clinical Workflow

Human-in-the-loop clinician sign-off workflow and multi-document longitudinal patient timelines.